

Regression Final Group Project_Group A

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1. Identify a data Table/Data Source that satisfies the requirements for your Project

Ans

- one dependent variable: Exam_Score
- 5 or more independent variables: Hours_Studied, Attendance, Extracurricular_Activities, Sleep_Hours, Motivation_Level, Tutoring_Sessions, School_Type
- two or more of the independent variables being categorical: Extracurricular_Activities, Motivation_Level, School_Type
- at least 100 observational rows: the data set has 6,607 observational rows

2. Describe your Data

Ans This dataset provides a comprehensive overview of various factors affecting student performance in exams. It includes information on study habits, attendance, parental involvement, and other aspects influencing academic success.

3. Use R code to import your Data Table into R Studio.

```
stu_perform <- read_csv("StudentPerformanceFactors.csv")
```

```
## Rows: 6607 Columns: 20
## -- Column specification -----
## Delimiter: ","
## chr (13): Parental_Involvement, Access_to_Resources, Extracurricular_Activit...
## dbl (7): Hours_Studied, Attendance, Sleep_Hours, Previous_Scores, Tutoring...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
stu_perform
```

```
## # A tibble: 6,607 x 20
##   Hours_Studied Attendance Parental_Involvement Access_to_Resources
##           <dbl>         <dbl> <chr>                <chr>
## 1             23           84 Low                    High
```

```
## 2      19      64 Low      Medium
## 3      24      98 Medium    Medium
## 4      29      89 Low      Medium
## 5      19      92 Medium    Medium
## 6      19      88 Medium    Medium
## 7      29      84 Medium    Low
## 8      25      78 Low      High
## 9      17      94 Medium    High
## 10     23      98 Medium    Medium
## # i 6,597 more rows
## # i 16 more variables: Extracurricular_Activities <chr>, Sleep_Hours <dbl>,
## #   Previous_Scores <dbl>, Motivation_Level <chr>, Internet_Access <chr>,
## #   Tutoring_Sessions <dbl>, Family_Income <chr>, Teacher_Quality <chr>,
## #   School_Type <chr>, Peer_Influence <chr>, Physical_Activity <dbl>,
## #   Learning_Disabilities <chr>, Parental_Education_Level <chr>,
## #   Distance_from_Home <chr>, Gender <chr>, Exam_Score <dbl>
```

4. Define/Describe all Variables of your Data Table

Ans

- Exam_Score: Final exam score.
- Hours_Studied: Number of hours spent studying per week.
- Attendance: Percentage of classes attended.
- Extracurricular_Activities: Participation in extracurricular activities (Yes, No).
- Sleep_Hours: Average number of hours of sleep per night.
- Motivation_Level: Student's level of motivation (Low, Medium, High).
- Tutoring_Sessions: Number of tutoring sessions attended per month.
- School_Type: Type of school attended (Public, Private).

5. Indicate source of your Data Table

Ans

Student Performance Factors: This data set is a synthetic dataset generated for educational and analytical purposes. The data is not sourced from any real-world institutions but is created to simulate realistic scenarios for analyzing student performance factors.

6. Missing value analysis

```
stu_perform <- stu_perform |>
  select(Exam_Score, Hours_Studied, Attendance, Extracurricular_Activities,
         Sleep_Hours, Motivation_Level, Tutoring_Sessions, School_Type)
```

```
sum(is.na(stu_perform))
```

```
## [1] 0
```

```
colSums(is.na(stu_perform))
```

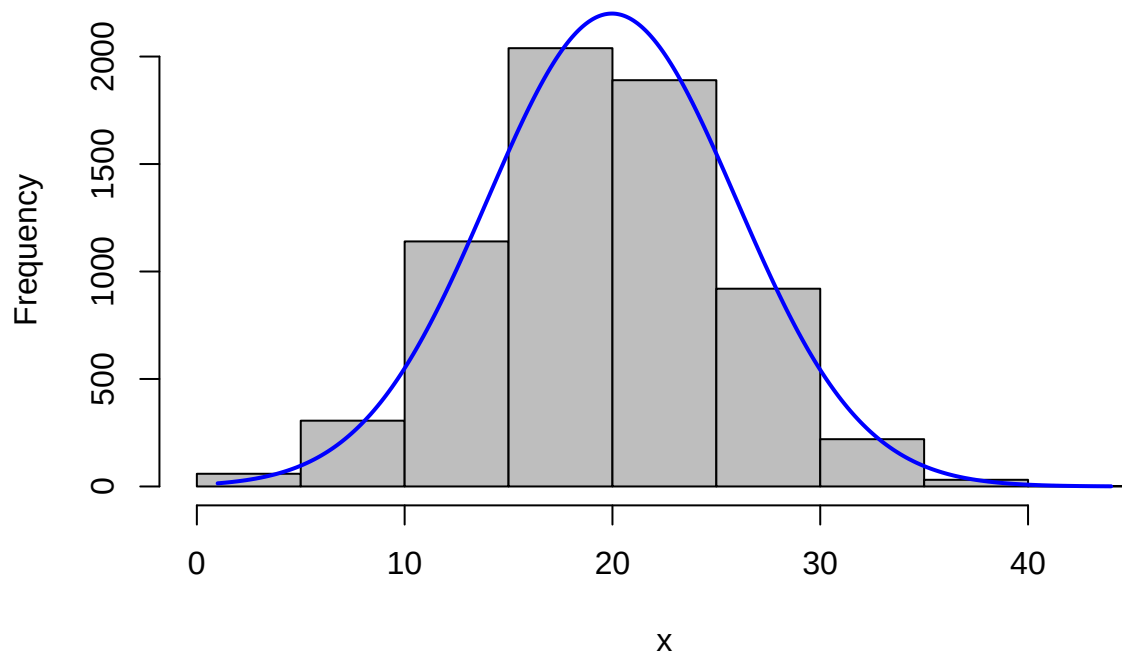
```
##           Exam_Score           Hours_Studied
##           0           0
##           Attendance Extracurricular_Activities
##           0           0
##           Sleep_Hours           Motivation_Level
##           0           0
##           Tutoring_Sessions           School_Type
##           0           0
```

There is no missing value in the data set.

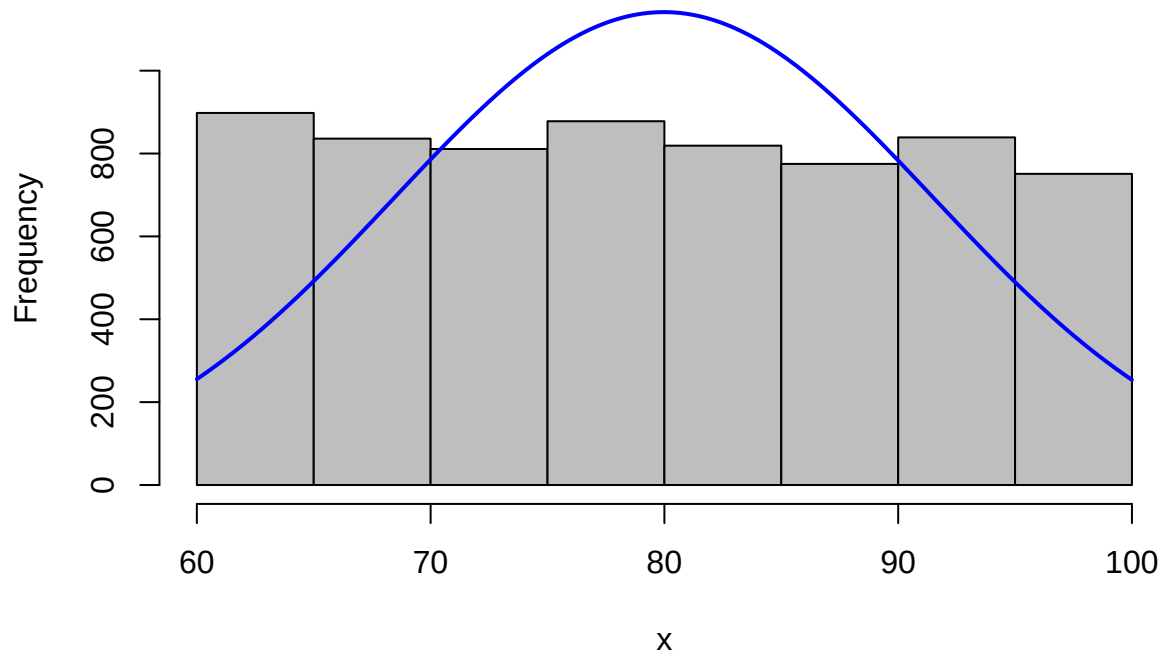
7. Generate relevant data visual plots that explore multicollinearity for the quantitative variables and normality for the quantitative variables as well. Also, use R code to confirm the levels of the categorical variables. (Pin Tzu)

```
#normality for the quantitative variables
```

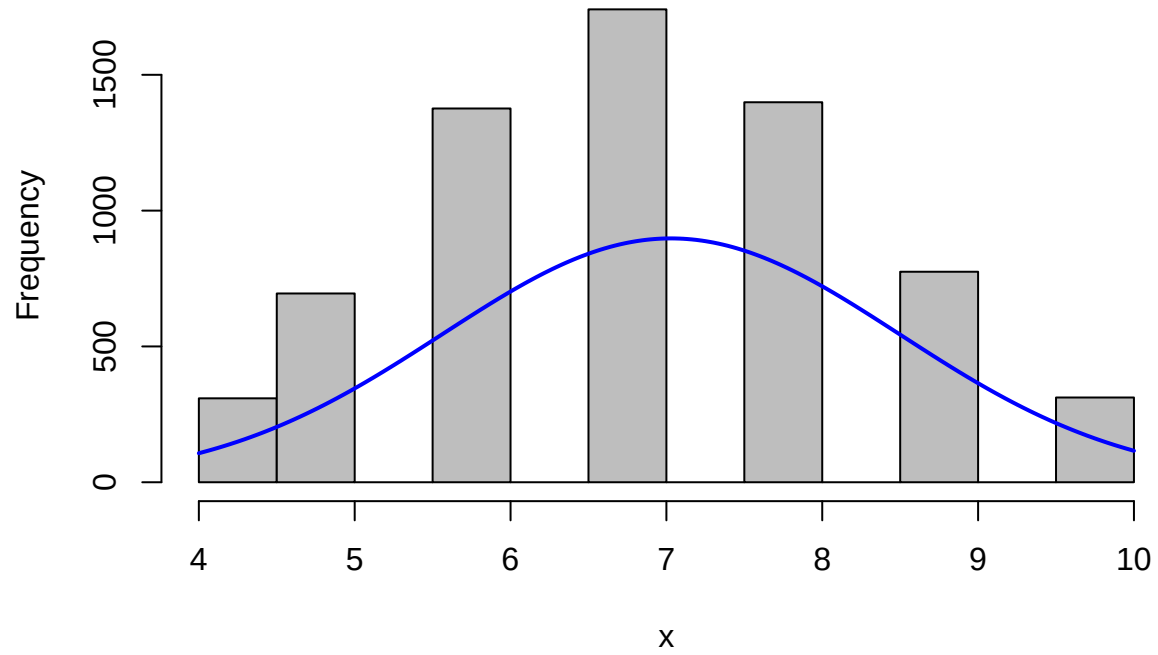
```
plotNormalHistogram(stu_perform$Hours_Studied)
```



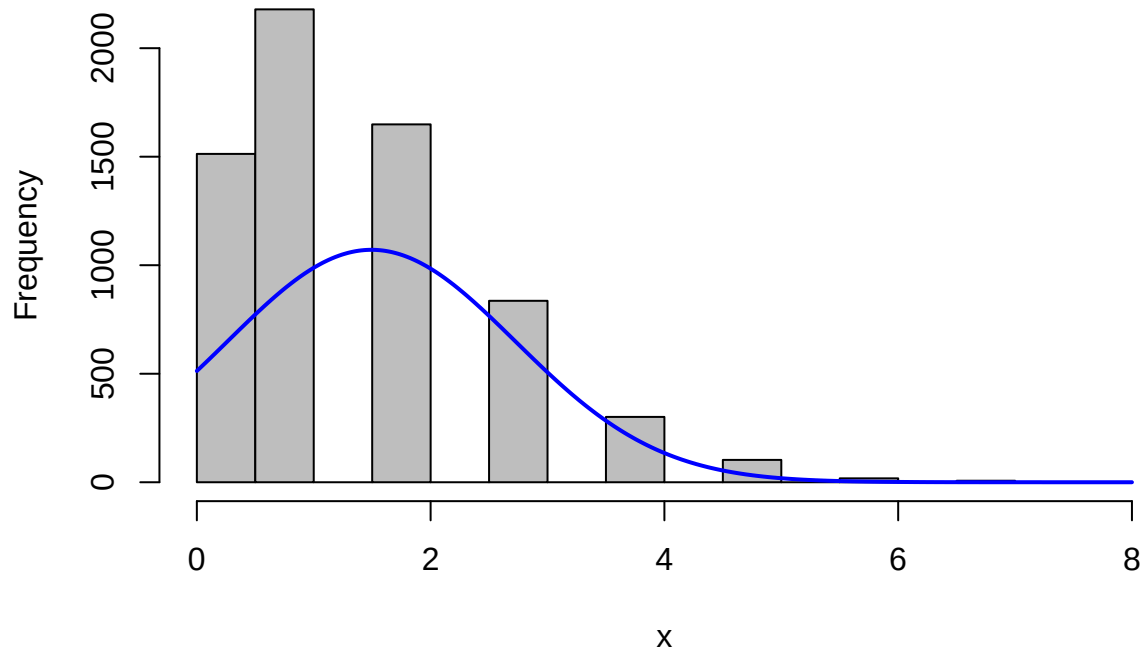
```
plotNormalHistogram(stu_perform$Attendance)
```



```
plotNormalHistogram(stu_perform$Sleep_Hours)
```



```
plotNormalHistogram(stu_perform$Tutoring_Sessions)
```



As these histogram, we can see the variable Hours_Studied, Attendance, and Sleep_Hours have normal distribution, but variable Tutoring_Sessions has right skewed distribution.

```
#level of categorical variables
EA <- factor(stu_perform$Extracurricular_Activities)
levels(EA)
```

```
## [1] "No" "Yes"
```

```
ML <- factor(stu_perform$Motivation_Level)
levels(ML)
```

```
## [1] "High" "Low" "Medium"
```

```
ST <- factor(stu_perform$School_Type)
levels(ST)
```

```
## [1] "Private" "Public"
```

the level of Extracurricular_Activities is no and yes.

the level of Motivation_Level is high, low, and medium.

the level of School_Type is private and public.

```
#only quantitative variable multiple regression model
only_numva <-
  stu_perform |>
  select(Exam_Score, Hours_Studied, Attendance,
         Sleep_Hours, Tutoring_Sessions)
only_numva
```

```
## # A tibble: 6,607 x 5
##   Exam_Score Hours_Studied Attendance Sleep_Hours Tutoring_Sessions
##   <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1         67         23         84         7         0
## 2         61         19         64         8         2
## 3         74         24         98         7         2
## 4         71         29         89         8         1
## 5         70         19         92         6         3
## 6         71         19         88         8         3
## 7         67         29         84         7         1
## 8         66         25         78         6         1
## 9         69         17         94         6         0
## 10        72         23         98         8         0
## # i 6,597 more rows
```

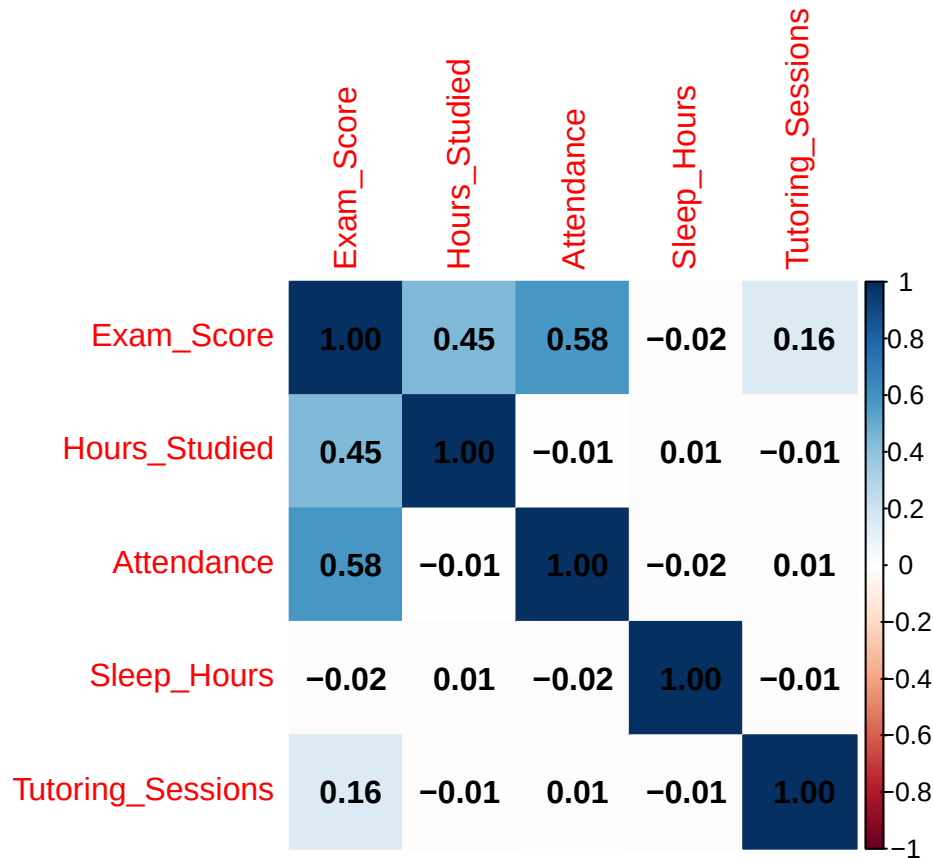
I selected the variables only included quantitative variables to test their multicollinearity.

```
#multicollinearity for quantitative variables
cor <- cor(only_numva)
cor
```

```
##           Exam_Score Hours_Studied Attendance Sleep_Hours
## Exam_Score      1.0000000    0.445454954  0.581071863 -0.01702163
## Hours_Studied   0.44545495    1.000000000 -0.009907859  0.01097669
## Attendance      0.58107186   -0.009907859  1.000000000 -0.01591783
## Sleep_Hours     -0.01702163    0.010976689 -0.015917826  1.00000000
## Tutoring_Sessions 0.15652519  -0.014282264  0.014323509 -0.01221611
##           Tutoring_Sessions
## Exam_Score      0.15652519
## Hours_Studied   -0.01428226
## Attendance      0.01432351
## Sleep_Hours     -0.01221611
## Tutoring_Sessions 1.00000000
```

They don't have multicollinearity problem, and I also create a multicollinearity plot.

```
#multicollinearity plot for quantitative variables
corrplot::corrplot(cor, method = "color", addCoef.col = "black")
```



8. Using R code, produce a full Regression Model that consists of quantitative and categorical variables. (Pin Tzu)

```
#full multiple regression model (include all variables)
all_lm <- lm(data = stu_perform, Exam_Score ~ Hours_Studied +
             Attendance+ Extracurricular_Activities + Sleep_Hours +
             Motivation_Level + Tutoring_Sessions + School_Type)
all_lm

##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities +
##     Sleep_Hours + Motivation_Level + Tutoring_Sessions + School_Type,
##     data = stu_perform)
##
## Coefficients:
##              (Intercept)              Hours_Studied
##              45.347739              0.295459
##              Attendance Extracurricular_ActivitiesYes
##              0.196689              0.526723
##              Sleep_Hours      Motivation_LevelLow
##              -0.029212              -1.045458
##              Motivation_LevelMedium      Tutoring_Sessions
```



```
##                -0.530504                0.487480
##                School_TypePublic
##                -0.008441
```

Full model:

Exam_Score = 45.347739 + 0.295459 (Hours_Studied) + 0.196689(Attendance) + 0.526723(Extracurricular_ActivitiesYes)-0.029212(Sleep_Hours) + -1.045458(Motivation_LevelLow)-0.530504(Motivation_LevelMedium) + 0.487480(Tutoring_Sessions)-0.008441(School_TypePublic)

9. Make use of the R generated dummy variable matrices(Pin Tzu)

```
contrasts(as.factor(stu_perform$Extracurricular_Activities))
```

```
##      Yes
## No      0
## Yes     1
```

```
contrasts(as.factor(stu_perform$Motivation_Level))
```

```
##      Low Medium
## High    0      0
## Low     1      0
## Medium  0      1
```

```
contrasts(as.factor(stu_perform$School_Type))
```

```
##      Public
## Private    0
## Public     1
```

10. Using only the quantitative variables as predictors, produce a model using matrix methods. Also use matrix methods to find the fitted values and the residuals (Dongni)

```
quantitative_data <- stu_perform[, c("Exam_Score", "Hours_Studied", "Attendance", "Sleep_Hours", "Tutoring_Sessions")]
X <- as.matrix(cbind(1, quantitative_data[, -1]))
y <- as.matrix(quantitative_data$Exam_Score)
```

```
beta <- solve(t(X) %*% X) %*% t(X) %*% y
beta
```

```
##      [,1]
## 1      45.10999572
## Hours_Studied    0.29455346
## Attendance       0.19647899
## Sleep_Hours      -0.02869808
## Tutoring_Sessions 0.48850816
```

quantitative variables model:

$\text{Exam_Score} = 45.11 + 0.2945535(\text{Hours_Studied}) + 0.196479(\text{Attendance}) - 0.02869808(\text{Sleep_Hours}) + 0.4885082(\text{Tutoring_Sessions})$

```
#fitted value
y_hat <- X %*% beta
head(y_hat)
```

```
##           [,1]
## [1,] 68.18807
## [2,] 64.02860
## [3,] 72.21035
## [4,] 71.39760
## [5,] 70.07591
## [6,] 69.23260
```

```
# residuals
residuals <- y - y_hat
head(residuals)
```

```
##           [,1]
## [1,] -1.18807345
## [2,] -3.02859815
## [3,]  1.78965097
## [4,] -0.39759920
## [5,] -0.07591407
## [6,]  1.76739803
```

11. Produce an output summary table to be used to analyze and evaluate the full model (Adjusted R squared, Standard Error, Significance of Variables, ect...) (Meng Hsuan)

```
summary(all_lm)
```

```
##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities +
##      Sleep_Hours + Motivation_Level + Tutoring_Sessions + School_Type,
##      data = stu_perform)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8598 -1.2017 -0.1610  0.8929 31.3856
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    45.347739   0.300547  150.884 < 2e-16 ***
## Hours_Studied    0.295459   0.005192   56.901 < 2e-16 ***
## Attendance       0.196689   0.002694   73.012 < 2e-16 ***
## Extracurricular_ActivitiesYes 0.526723   0.063368    8.312 < 2e-16 ***
```

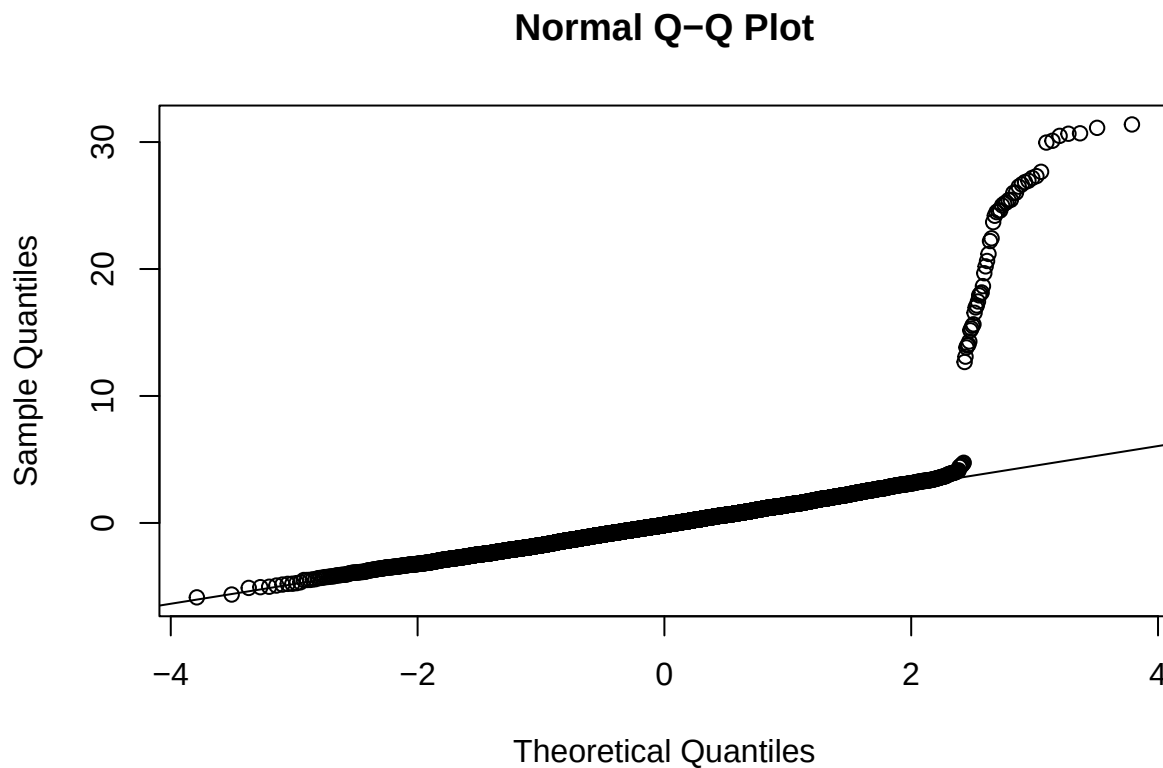
```
## Sleep_Hours          -0.029212   0.021183  -1.379    0.168
## Motivation_LevelLow   -1.045458   0.090242 -11.585 < 2e-16 ***
## Motivation_LevelMedium -0.530504   0.082180  -6.455 1.16e-10 ***
## Tutoring_Sessions     0.487480   0.025276  19.286 < 2e-16 ***
## School_TypePublic     -0.008441   0.067605  -0.125    0.901
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.527 on 6598 degrees of freedom
## Multiple R-squared:  0.5786, Adjusted R-squared:  0.5781
## F-statistic: 1132 on 8 and 6598 DF, p-value: < 2.2e-16
```

According to the summary table of full model, it shows that the p value of variable “Sleep_Hours” and “School_TypePublic” is higher than 0.05 which is not significant. Although Tutoring_Sessions is not normality, but its p value show it is significant, so we will consider it in our final model.

Otherwise, the Adjusted R-squared is 0.5781 which indicate approximately half of the observed variation can be explained by the model’s inputs.

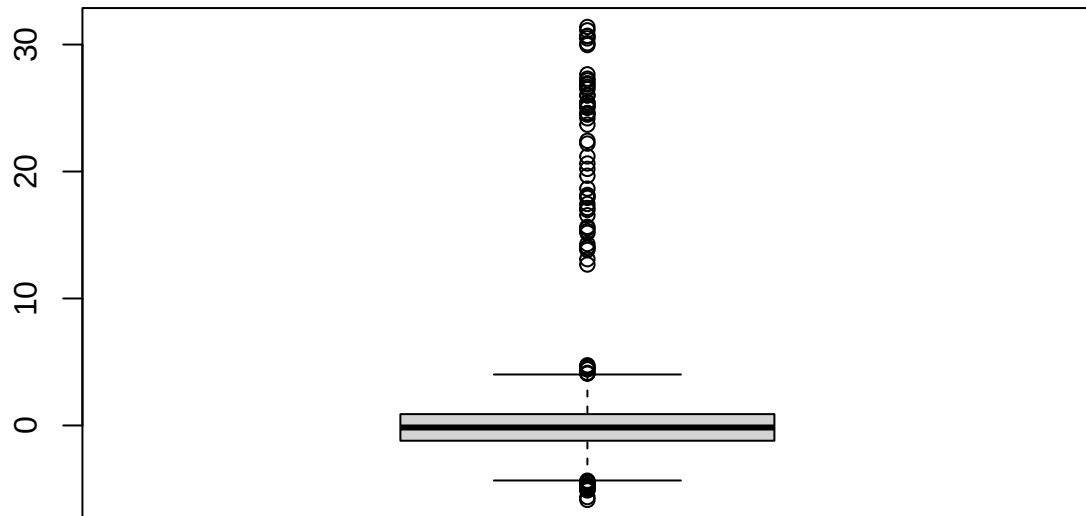
Also, residual standard error is 2.527 unit far from the regression line in average.

```
qqnorm(all_lm$residuals)
qqline(all_lm$residuals)
```



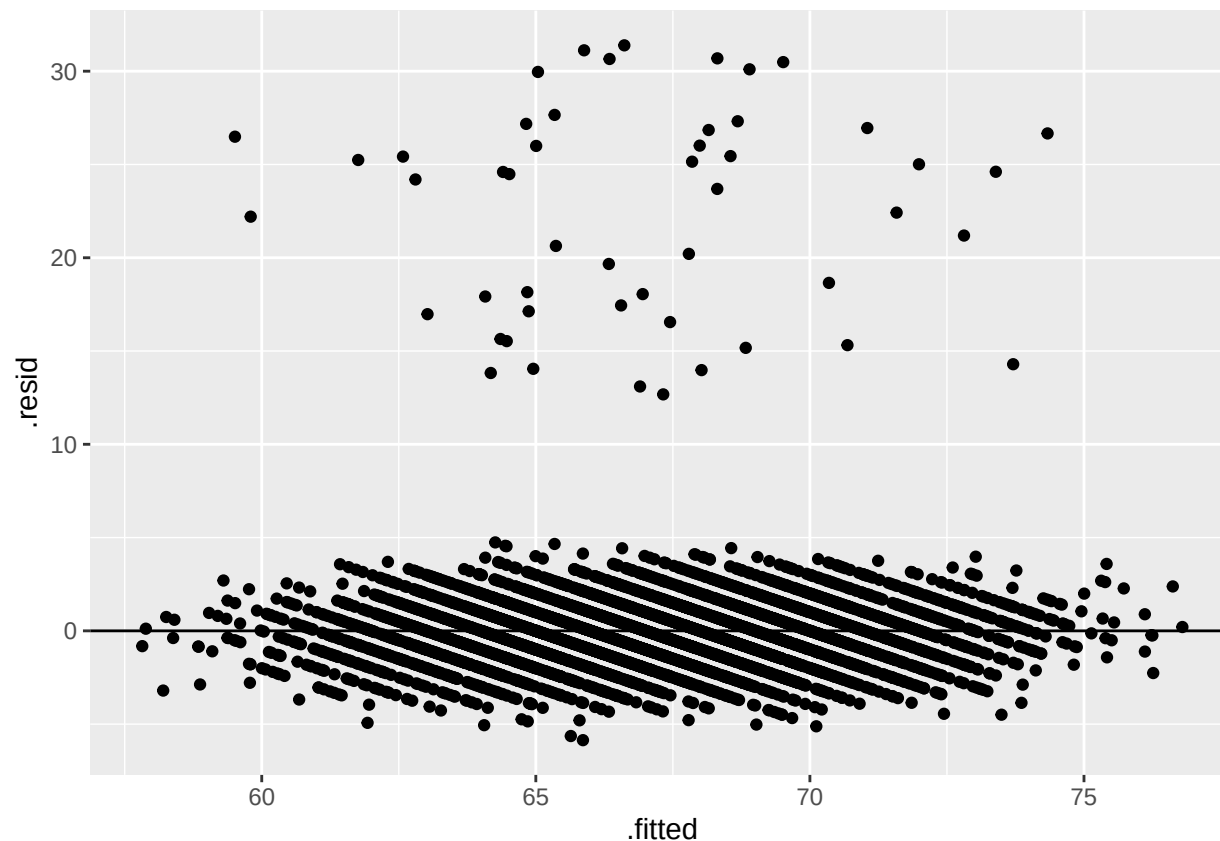
We used the residual plot to see the normality. The data points do not always align in the middle of the plot. The distribution of the residuals is not pretty normal. There are lots of outliers.

```
boxplot(all_lm$residuals)
```



The residual boxplot also presents many outliers.

```
ggplot(all_lm, aes(x = .fitted, y = .resid)) +  
  geom_point() +  
  geom_hline(yintercept = 0)
```



In this residual plot doesn't suggest a linear relationship which may caused by many outliers.
We conclude that the data set should be transformed to exclude the outliers.

```
ex_outlier <- stu_perform |>
  mutate(
    residuals = all_lm$residuals) |>
  filter(residuals <10)
ex_outlier
```

```
## # A tibble: 6,557 x 9
##   Exam_Score Hours_Studied Attendance Extracurricular_Activities Sleep_Hours
##   <dbl>      <dbl>      <dbl> <chr>                                <dbl>
## 1         67         23         84 No                                  7
## 2         61         19         64 No                                  8
## 3         74         24         98 Yes                                7
## 4         71         29         89 Yes                                8
## 5         70         19         92 Yes                                6
## 6         71         19         88 Yes                                8
## 7         67         29         84 Yes                                7
## 8         66         25         78 Yes                                6
## 9         69         17         94 No                                  6
## 10        72         23         98 Yes                                8
## # i 6,547 more rows
## # i 4 more variables: Motivation_Level <chr>, Tutoring_Sessions <dbl>,
## #   School_Type <chr>, residuals <dbl>
```

```

ex_outlier_lm <- lm(data = ex_outlier, Exam_Score ~ Hours_Studied +
                    Attendance+ Extracurricular_Activities + Sleep_Hours +
                    Motivation_Level + Tutoring_Sessions + School_Type)
summary(ex_outlier_lm)

##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities +
##     Sleep_Hours + Motivation_Level + Tutoring_Sessions + School_Type,
##     data = ex_outlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.6022 -1.0397 -0.0042  1.0285  4.8638
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    44.921899   0.184071  244.046 <2e-16 ***
## Hours_Studied     0.299047   0.003178   94.089 <2e-16 ***
## Attendance        0.197508   0.001648  119.872 <2e-16 ***
## Extracurricular_ActivitiesYes 0.471208   0.038756   12.158 <2e-16 ***
## Sleep_Hours      -0.009896   0.012961   -0.763    0.445
## Motivation_LevelLow -1.002195   0.055232  -18.145 <2e-16 ***
## Motivation_LevelMedium -0.491789   0.050312   -9.775 <2e-16 ***
## Tutoring_Sessions    0.489837   0.015459   31.686 <2e-16 ***
## School_TypePublic   -0.036965   0.041340   -0.894    0.371
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.54 on 6548 degrees of freedom
## Multiple R-squared:  0.7886, Adjusted R-squared:  0.7883
## F-statistic: 3053 on 8 and 6548 DF, p-value: < 2.2e-16

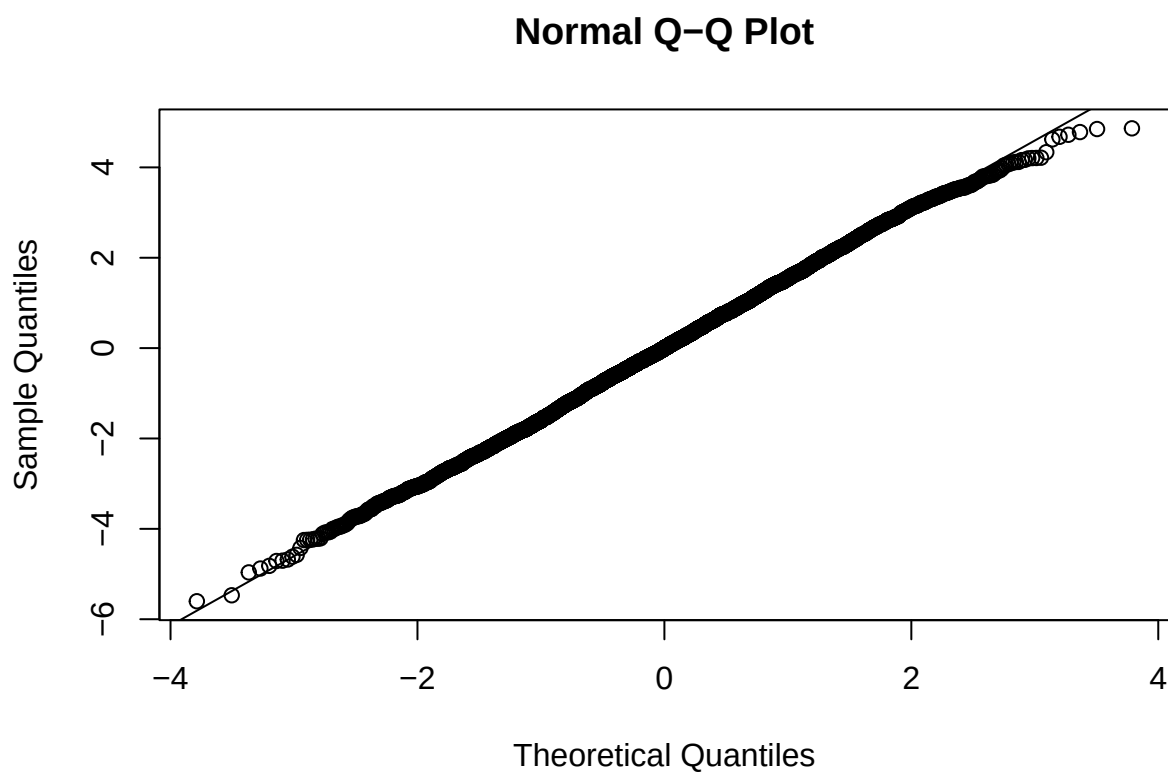
```

After excluding the outliers from the full model, the residual standard error gets smaller and Adjusted R-squared is higher to get close to one which means the full model gets better after transformation.

```

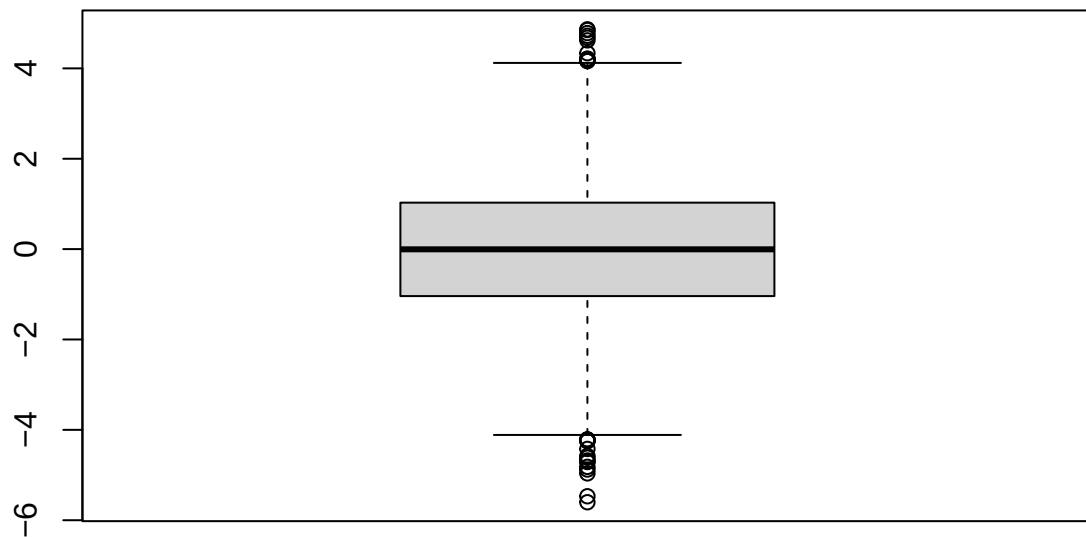
qqnorm(ex_outlier_lm$residuals)
qqline(ex_outlier_lm$residuals)

```



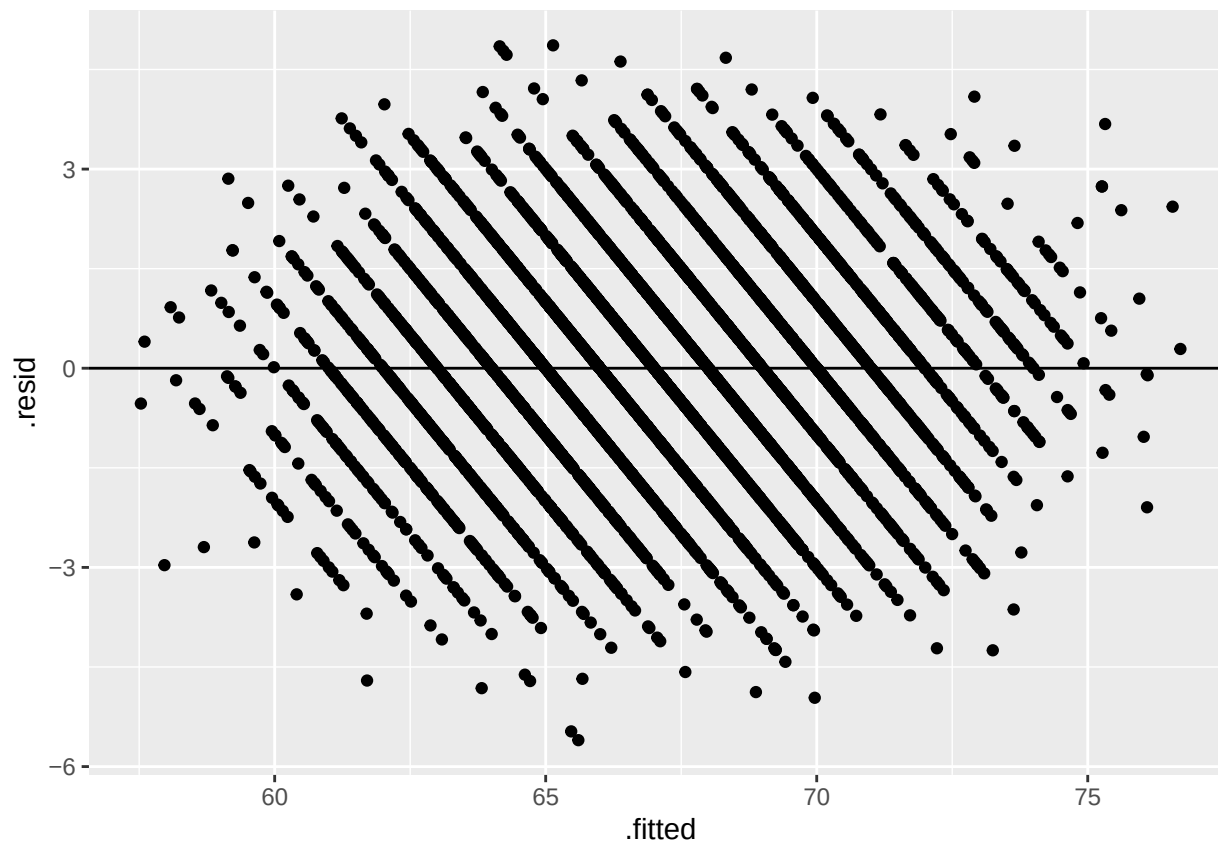
The residual plot shows the data points align on the line after excluding the outliers.

```
boxplot(ex_outlier_lm$residuals)
```



The outlier is fewer in the boxplot after excluding the outliers.

```
ggplot(ex_outlier_lm, aes(x = .fitted, y = .resid)) +  
  geom_point() +  
  geom_hline(yintercept = 0)
```

After excluding the outliers, the residual plot suggests a linear relationship because it shows the constant variance.

12. Use procedures and techniques explored in class to produce confidence intervals for the independent quantitative variables of your model. Choose at least two of the quantitative variables to find confidence intervals for.(Dongni)

```
model <- lm(Exam_Score ~ Hours_Studied + Sleep_Hours + Attendance +
            Tutoring_Sessions, data = stu_perform)

confint(model, level = 0.95)
```

```
##              2.5 %      97.5 %
## (Intercept)  44.53714961 45.68284184
## Hours_Studied  0.28422073 0.30488618
## Sleep_Hours   -0.07086236 0.01346619
## Attendance    0.19111822 0.20183975
## Tutoring_Sessions 0.43820361 0.53881272
```

13. Produce a reduced model (removing variables of your choice with justification). Use R summary coding for both models and offer justification for choosing one model over the other. Also apply this model technique: Mallows's Cp&AIC&SBC with `ols_step_best_subset()` function from 11/20 model selection.(Meng Hsuan)

Due to the not significant variable "Sleep_Hours" and "School_TypePublic", we decided to remove them from our model.

Firstly, we examine the model with full data instead of excluding the outliers.

```
stu_perform_lm2 <- lm(Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities + Motivation,
summary(stu_perform_lm2)
```

```
##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities +
##      Motivation_Level + Tutoring_Sessions, data = stu_perform)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8246 -1.2040 -0.1659  0.9010 31.3539
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    45.132305   0.254494  177.342 < 2e-16 ***
## Hours_Studied     0.295382   0.005192   56.890 < 2e-16 ***
## Attendance       0.196754   0.002693   73.058 < 2e-16 ***
## Extracurricular_ActivitiesYes 0.526556   0.063367    8.310 < 2e-16 ***
## Motivation_LevelLow -1.045182   0.090233  -11.583 < 2e-16 ***
## Motivation_LevelMedium -0.530397   0.082177   -6.454 1.16e-10 ***
## Tutoring_Sessions  0.487866   0.025273   19.304 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.527 on 6600 degrees of freedom
## Multiple R-squared:  0.5785, Adjusted R-squared:  0.5781
## F-statistic: 1510 on 6 and 6600 DF, p-value: < 2.2e-16
```

Still, the Adjusted R-squared shows as same.

We used stepwise regression method to choose the best model to predict student exam score.

```
intercept_only <- lm(Exam_Score ~ 1, data=stu_perform)
intercept_only
```

```
##
## Call:
## lm(formula = Exam_Score ~ 1, data = stu_perform)
##
## Coefficients:
## (Intercept)
##      67.24
```

```
all <- lm(Exam_Score ~ ., data=stu_perform)
all
```

```
##
## Call:
## lm(formula = Exam_Score ~ ., data = stu_perform)
##
## Coefficients:
##              (Intercept)              Hours_Studied
##              45.347739              0.295459
##              Attendance  Extracurricular_ActivitiesYes
##              0.196689              0.526723
##              Sleep_Hours              Motivation_LevelLow
##              -0.029212              -1.045458
##              Motivation_LevelMedium  Tutoring_Sessions
##              -0.530504              0.487480
##              School_TypePublic
##              -0.008441
```

```
forward <- step(intercept_only, direction='forward',
                scope=formula(all), trace=0)
forward
```

```
##
## Call:
## lm(formula = Exam_Score ~ Attendance + Hours_Studied + Tutoring_Sessions +
##      Motivation_Level + Extracurricular_Activities, data = stu_perform)
##
## Coefficients:
##              (Intercept)              Attendance
##              45.1323              0.1968
##              Hours_Studied              Tutoring_Sessions
##              0.2954              0.4879
##              Motivation_LevelLow  Motivation_LevelMedium
##              -1.0452              -0.5304
## Extracurricular_ActivitiesYes
##              0.5266
```

```
forward$anova
```

	Step	Df	Deviance	Resid. Df	Resid. Dev	AIC
## 1		NA	NA	6606	99986.08	17952.57
## 2	+ Attendance	-1	33759.7506	6605	66226.33	15232.79
## 3	+ Hours_Studied	-1	20358.4031	6604	45867.93	12807.96
## 4	+ Tutoring_Sessions	-1	2390.2298	6603	43477.70	12456.37
## 5	+ Motivation_Level	-2	890.8658	6601	42586.83	12323.59
## 6	+ Extracurricular_Activities	-1	440.9337	6600	42145.90	12256.82

We found out the best model is:

Exam_Score = 45.1323 + 0.1968(Attendance) + 0.2954(Hours_Studied) + 0.4879(Tutoring_Sessions) - 1.0452(Motivation_LevelLow) - 0.5304(Motivation_LevelMedium) + 0.5266(Extracurricular_ActivitiesYes)

```
stu_perform_lm3 <- lm(Exam_Score ~ Attendance + Hours_Studied +
  Tutoring_Sessions + Motivation_Level + Extracurricular_Activities,
  data = stu_perform)

ols_step_best_subset(stu_perform_lm3)
```

```
##                                     Best Subsets Regression
## -----
## Model Index    Predictors
## -----
##      1         Attendance
##      2         Attendance Hours_Studied
##      3         Attendance Hours_Studied Tutoring_Sessions
##      4         Attendance Hours_Studied Tutoring_Sessions Motivation_Level
##      5         Attendance Hours_Studied Tutoring_Sessions Motivation_Level Extracurricular_Activities
## -----
##
##                                     Subsets Regression Summary
## -----
## Model      R-Square    Adj.      Pred      C(p)      AIC      SBIC      SBC
## -----
##      1         0.3376    0.3375    0.3372    3767.9687    33984.6478    15233.0789    34005.0355
##      2         0.5413    0.5411    0.5408    581.8656    31559.8173    12809.4697    31587.0009
##      3         0.5652    0.5650    0.5646    209.5583    31208.2238    12458.1314    31242.2032
##      4         0.5741    0.5737    0.5733     74.0497    31075.4390    12323.4920    31123.0102
##      5         0.5785    0.5781    0.5776      7.0000    31008.6751    12256.8340    31063.0422
## -----
## AIC: Akaike Information Criteria
## SBIC: Sawa's Bayesian Information Criteria
## SBC: Schwarz Bayesian Criteria
## MSEP: Estimated error of prediction, assuming multivariate normality
## FPE: Final Prediction Error
## HSP: Hocking's Sp
## APC: Amemiya Prediction Criteria
```

According to the table, the fifth model including independent variable of Attendance, Hours_Studied, Tutoring_Sessions, Motivation_Level, and Extracurricular_Activities has highest Adjusted R-squared and lowest AIC, so that one is the best model.

In the second place, we examine the model without outliers.

```
ex_outlier_lm2 <- lm(Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities + Motivation_Level,
  data = ex_outlier)

summary(ex_outlier_lm2)
```

```
##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Attendance + Extracurricular_Activities +
##      Motivation_Level + Tutoring_Sessions, data = ex_outlier)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -5.5662 -1.0459 -0.0056 1.0271 4.8752
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    44.823012   0.155904  287.503   <2e-16 ***
## Hours_Studied    0.299031   0.003178   94.093   <2e-16 ***
## Attendance       0.197554   0.001647  119.940   <2e-16 ***
## Extracurricular_ActivitiesYes 0.470948   0.038753   12.152   <2e-16 ***
## Motivation_LevelLow -1.001522   0.055225  -18.135   <2e-16 ***
## Motivation_LevelMedium -0.491439   0.050308   -9.769   <2e-16 ***
## Tutoring_Sessions  0.489839   0.015457   31.691   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.54 on 6550 degrees of freedom
## Multiple R-squared:  0.7886, Adjusted R-squared:  0.7884
## F-statistic: 4071 on 6 and 6550 DF, p-value: < 2.2e-16
```

The residual standard error is smaller and Adjusted R-squared is higher compare to the model with full data.

We used stepwise regression method to choose the best model after transformaion to predict student exam score.

```
ex_outlier <- ex_outlier |>
  select(Exam_Score,Hours_Studied,Attendance, Extracurricular_Activities,
    Motivation_Level, Tutoring_Sessions)

intercept_only2 <- lm(Exam_Score ~ 1, data = ex_outlier)
intercept_only2
```

```
##
## Call:
## lm(formula = Exam_Score ~ 1, data = ex_outlier)
##
## Coefficients:
## (Intercept)
##      67.07
```

```
all2 <- lm(Exam_Score ~ ., data = ex_outlier)
all2
```

```
##
## Call:
## lm(formula = Exam_Score ~ ., data = ex_outlier)
##
## Coefficients:
##              (Intercept)              Hours_Studied
##              44.823012              0.2990
##              Attendance Extracurricular_ActivitiesYes
##              0.1976              0.4709
##              Motivation_LevelLow Motivation_LevelMedium
##              -1.0015              -0.4914
```

```
##           Tutoring_Sessions
##           0.4898
```

```
forward2 <- step(intercept_only, direction='forward',
                 scope=formula(all2), trace=0)
forward2
```

```
##
## Call:
## lm(formula = Exam_Score ~ Attendance + Hours_Studied + Tutoring_Sessions +
##     Motivation_Level + Extracurricular_Activities, data = stu_perform)
##
## Coefficients:
##              (Intercept)              Attendance
##              45.1323              0.1968
##              Hours_Studied              Tutoring_Sessions
##              0.2954              0.4879
##              Motivation_LevelLow      Motivation_LevelMedium
##              -1.0452              -0.5304
## Extracurricular_ActivitiesYes
##              0.5266
```

```
forward2$anova
```

```
##              Step Df    Deviance Resid. Df Resid. Dev      AIC
## 1              NA      NA          6606    99986.08 17952.57
## 2      + Attendance -1  33759.7506          6605    66226.33 15232.79
## 3      + Hours_Studied -1  20358.4031          6604    45867.93 12807.96
## 4      + Tutoring_Sessions -1  2390.2298          6603    43477.70 12456.37
## 5      + Motivation_Level -2    890.8658          6601    42586.83 12323.59
## 6 + Extracurricular_Activities -1    440.9337          6600    42145.90 12256.82
```

We found out the best model after transformation is:

Exam_Score = 45.1323 + 0.1968(Attendance) + 0.2954(Hours_Studied) + 0.4879(Tutoring_Sessions) - 1.0452(Motivation_LevelLow) -0.5304(Motivation_LevelMedium) + 0.5266(Extracurricular_ActivitiesYes)

```
ex_outlier_lm3 <- lm(Exam_Score ~ Attendance + Hours_Studied +
  Tutoring_Sessions + Motivation_Level + Extracurricular_Activities,
  data = ex_outlier)

ols_step_best_subset(ex_outlier_lm3)
```

```
##                                     Best Subsets Regression
## -----
## Model Index    Predictors
## -----
##      1      Attendance
##      2      Attendance Hours_Studied
##      3      Attendance Hours_Studied Tutoring_Sessions
##      4      Attendance Hours_Studied Tutoring_Sessions Motivation_Level
##      5      Attendance Hours_Studied Tutoring_Sessions Motivation_Level Extracurricular_Activities
```

```
## -----
##
##                               Subsets Regression Summary
## -----
##      Adj.      Pred
## Model  R-Square R-Square R-Square    C(p)      AIC      SBIC      SBC
## -----
##    1      0.4583    0.4582    0.4579  10227.8512  30438.2631  11827.1237  30458.6279
##    2      0.7402    0.7401    0.7399   1497.8290  25622.7577   7013.6170  25649.9108
##    3      0.7727    0.7726    0.7724    491.6651  24747.2740   6138.7550  24781.2154
##    4      0.7838    0.7836    0.7834    152.6812  24423.8091   5813.6382  24471.3271
##    5      0.7886    0.7884    0.7881     7.0000  24279.6120   5669.6649  24333.9183
## -----
## AIC: Akaike Information Criteria
## SBIC: Sawa's Bayesian Information Criteria
## SBC: Schwarz Bayesian Criteria
## MSE: Estimated error of prediction, assuming multivariate normality
## FPE: Final Prediction Error
## HSP: Hocking's Sp
## APC: Amemiya Prediction Criteria
```

According to the table, the fifth model including independent variable of Attendance, Hours_Studied, Tutoring_Sessions, Motivation_Level, and Extracurricular_Activities has highest Adjusted R-squared and lowest AIC, so that one is the best model after excluding the outliers.

14. Research and apply a model analysis technique not discussed in class to your full model or reduced model. Fully explain the technique or procedure and how it is being applied to your specific model. (Dongni)

```
student1 <- stu_perform |>
  select("Hours_Studied", "Attendance", "Sleep_Hours", "Tutoring_Sessions", "Motivation_Level", "Extracurricular_Activities")
```

Subset Selection, specifically implemented through the regsubsets function, uses Exhaustive Search to identify the combinations of independent variables that best explain the target variable (Exam_Score). It evaluates all possible combinations of predictors and uses statistical metrics, such as Adjusted R-squared (Adj.R2), Mallows' Cp (Cp), and Bayesian Information Criterion (BIC), to assess each model's performance and assist in selecting the optimal model.

```
model_try <- regsubsets(Exam_Score ~ ., data = student1, method = "exhaustive")
summary(model_try)
```

```
## Subset selection object
## Call: regsubsets.formula(Exam_Score ~ ., data = student1, method = "exhaustive")
## 7 Variables (and intercept)
##
##      Forced in Forced out
## Hours_Studied      FALSE      FALSE
## Attendance         FALSE      FALSE
## Sleep_Hours        FALSE      FALSE
## Tutoring_Sessions  FALSE      FALSE
## Motivation_LevelLow FALSE      FALSE
## Motivation_LevelMedium FALSE      FALSE
```

```
## Extracurricular_ActivitiesYes    FALSE    FALSE
## 1 subsets of each size up to 7
## Selection Algorithm: exhaustive
##      Hours_Studied Attendance Sleep_Hours Tutoring_Sessions
## 1 ( 1 ) " "          "*"          " "          " "
## 2 ( 1 ) "*"          "*"          " "          " "
## 3 ( 1 ) "*"          "*"          " "          "*"
## 4 ( 1 ) "*"          "*"          " "          "*"
## 5 ( 1 ) "*"          "*"          " "          "*"
## 6 ( 1 ) "*"          "*"          " "          "*"
## 7 ( 1 ) "*"          "*"          "*"          "*"
##      Motivation_LevelLow Motivation_LevelMedium
## 1 ( 1 ) " "          " "
## 2 ( 1 ) " "          " "
## 3 ( 1 ) " "          " "
## 4 ( 1 ) "*"          " "
## 5 ( 1 ) "*"          " "
## 6 ( 1 ) "*"          "*"
## 7 ( 1 ) "*"          "*"
##      Extracurricular_ActivitiesYes
## 1 ( 1 ) " "
## 2 ( 1 ) " "
## 3 ( 1 ) " "
## 4 ( 1 ) " "
## 5 ( 1 ) "*"
## 6 ( 1 ) "*"
## 7 ( 1 ) "*"

```

```
res_sum <- summary(model_try)
data.frame(
  Adj.R2 = res_sum$adjr2,
  CP = res_sum$cp,
  BIC = res_sum$bic
)
```

```
##      Adj.R2      CP      BIC
## 1 0.3375442 3769.387368 -2704.181
## 2 0.5411180  582.848177 -5122.215
## 3 0.5649649  210.489667 -5467.013
## 4 0.5710824  115.718148 -5552.785
## 5 0.5755005   47.567057 -5613.398
## 6 0.5780992    7.902839 -5646.174
## 7 0.5781569    8.000000 -5639.283

```

The outcome shows the seventh model including independent variable of Hours_Studied, Attendance, Sleep_Hours, Tutoring_Sessions, Motivation_Level, and Extracurricular_Activities has highest Adjusted R-squared which indicates the best model.

```
coef(model_try,7)
```

```
##      (Intercept)      Hours_Studied
##      45.34139460      0.29545837
##      Attendance      Sleep_Hours

```


##	0.19669554	-0.02921849
##	Tutoring_Sessions	Motivation_LevelLow
##	0.48745279	-1.04530478
##	Motivation_LevelMedium	Extracurricular_ActivitiesYes
##	-0.53042024	0.52668052

The best model:

$$\text{Exam_Score} = 45.34139460 + 0.29545837(\text{Hours_Studied}) + 0.19669554(\text{Attendance}) - 0.02921849(\text{Sleep_Hours}) + 0.48745279(\text{Tutoring_Sessions}) - 1.04530478(\text{Motivation_LevelLow}) - 0.53042024(\text{Motivation_LevelMedium}) + 0.52668052(\text{Extracurricular_ActivitiesYes})$$

15. Offer final summary perspectives about the data and the models that you produce, suggesting how your models or model analysis enhanced your understanding of the data. (4 or 5 sentences) (ALL)

This study explores the factors influencing student exam performance using a dataset of 6,607 observations, with both quantitative and categorical variables. A full regression model revealed significant predictors like study hours, attendance, extracurricular participation, and tutoring sessions, while sleep hours and school type were found non-significant. Multicollinearity among quantitative variables was not a concern. We found out excluding outliers improved model fit, raising the adjusted R-squared and reducing residual error. Ultimately, regarding the result of stepwise regression method and subset selection, the best model to explain student's exam score should include the variable of Hours_Studied, Attendance, Sleep_Hours, Tutoring_Sessions, Motivation_Level, and Extracurricular_Activities.